

Final Project

Amazon Reviews Light

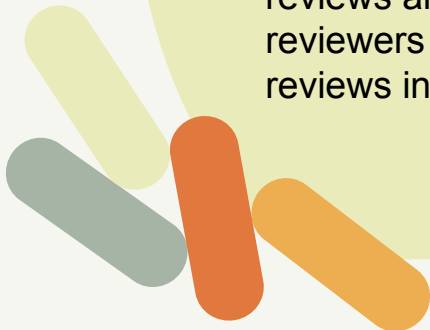
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December 12th, 2025

Executive Summary with meaningful insights

Our cleaned dataset of over 55 million Amazon reviews shows that a very small group of “super-reviewers” drives a huge share of activity: the top reviewer alone has 2,836 posts, and the top 20 all exceed 1,200. Most reviews come from categories like Beauty, Cell Phones & Accessories, and Automotive. Products with the most reviews aren’t the highest-rated: volume tends to cluster around 4.3–4.6 stars, while near-perfect 4.9–5.0 items are usually niche or newer. Price doesn’t meaningfully affect satisfaction, expensive products simply get fewer reviews. Review language also tells a clear story: 5-star comments focus on ease and quality, while 1-star reviews highlight problems and frustration. Older reviews show more copy-paste patterns, but recent reviews are more varied, possibly due to stricter moderation. Finally, top reviewers tend to specialize, each of the top five writes most of their reviews in just one category.



Source Dataset Overview (Raw data overview) & Methodology

Reviews Dataset

- Very large: ~65 million customer reviews
- Each row represents one review by one user for one product
- Key fields include:
 - Product ID (asin)
 - User ID (user_id)
 - Rating (1–5)
 - Review text + title
 - Timestamp
 - Purchase metadata (verified_purchase, helpful_vote)
- Most important fields (rating, text, product ID) are well populated
- A few fields have inconsistent or missing values (e.g., timestamps in seconds vs milliseconds)

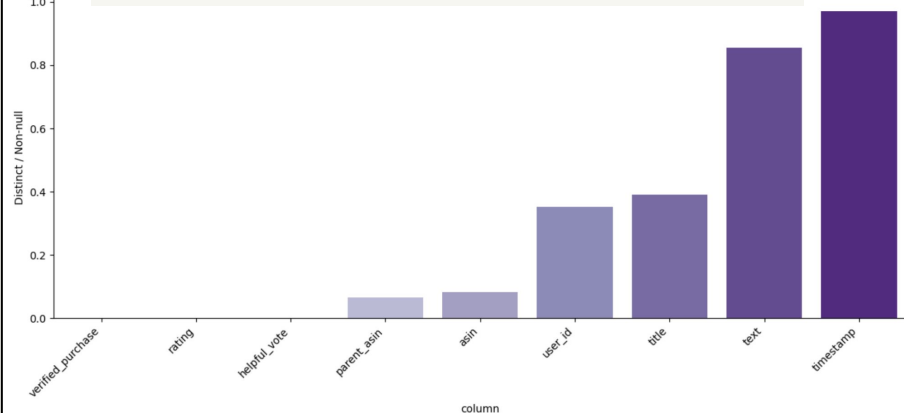
Meta Dataset

- Contains ~4.3 million Amazon products
- Includes:
 - Product name (title)
 - Parent ASIN
 - Price
 - Categories (nested list)
 - Description
- Review summary fields (average_rating, rating_number)
- Some columns are almost always missing, such as author, subtitle, bought_together have >99% missing
- Price and category fields vary widely in completeness and structure

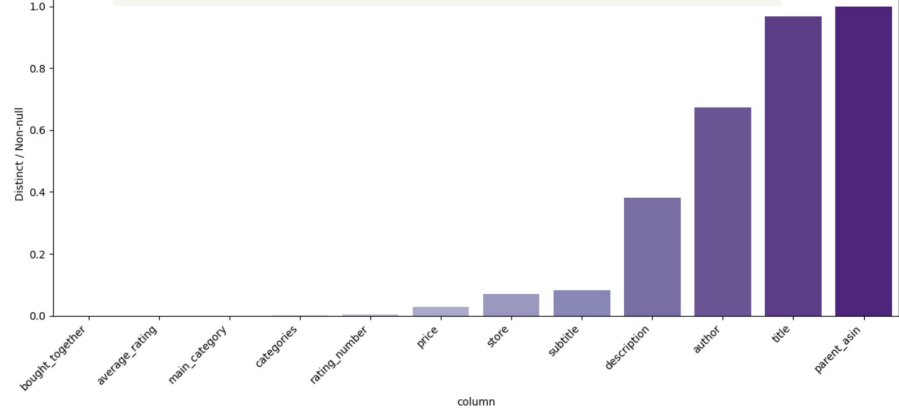
Summary of the raw datasets

- Several metadata fields are either barely populated or almost identical across rows
- Deeply nested structures (like categories) make processing harder
- Certain fields don't help analysis because they lack variation or are mostly empty
- Some numeric fields contain invalid or extreme values
- The scale requires careful filtering before running similarity analysis
- Therefore Data cleaning is necessary

Low Information Columns for the Reviews Dataset



Low Information Columns for the Meta Dataset



Data cleaning methodology and filtering approach

Review dataset cleaning methodology

Filtered valid records

- Kept only ratings 1–5
- Dropped rows missing asin, user_id, or text

Text quality filtering

- Removed extremely short or corrupted reviews
- Limited text length to avoid garbage or log-like content

Timestamp standardization

- Converted mixed formats (seconds vs. milliseconds)
- Removed impossible / future timestamps

Added analytic fields

- review_date, review_year, review_month

Meta dataset cleaning methodology

Flattened nested fields

- Extracted author.name
- Cleaned price strings (remove \$/, → numeric price)

Category handling

- Selected top-level category from nested lists
- Preserved full category hierarchy (categories_raw) if needed

Removed low-quality rows

- Dropped missing parent_asin
- Filtered invalid or extreme price values

Deduplicated products

Ensured one row per parent_asin

Reviews and Meta dataset Join and Final Prep

Left-join on parent_asin

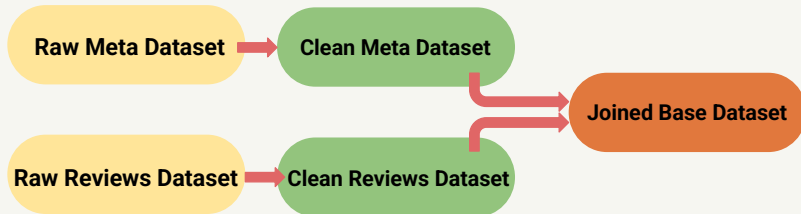
Explicit column selection to avoid nested noise

Created unified product_id

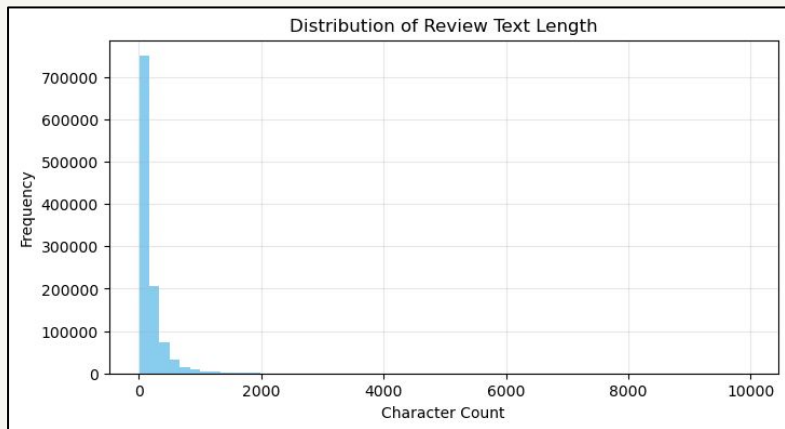
Post-join refinement

- Removed duplicates (user_id, asin, text)
- Normalized review text (lowercase + regex cleaning)
- Dropped reviews with blank normalized text
- Required products to have ≥ 5 reviews
- Added price_clean and filled missing values

Cleaned dataset overview



Final dataset contains 55M reviews across 24 structured fields after removing invalid timestamps, missing product IDs, empty review text, and duplicate reviews.

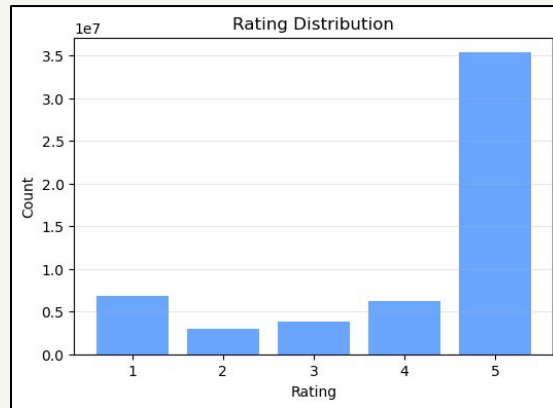


Most reviews are short-form comments, typically between 20 and 200 characters, indicating high user activity but concise expression. A long tail of very lengthy reviews (1,000+ characters) appears, but these are rare relative to the full dataset.

Top Category Distribution

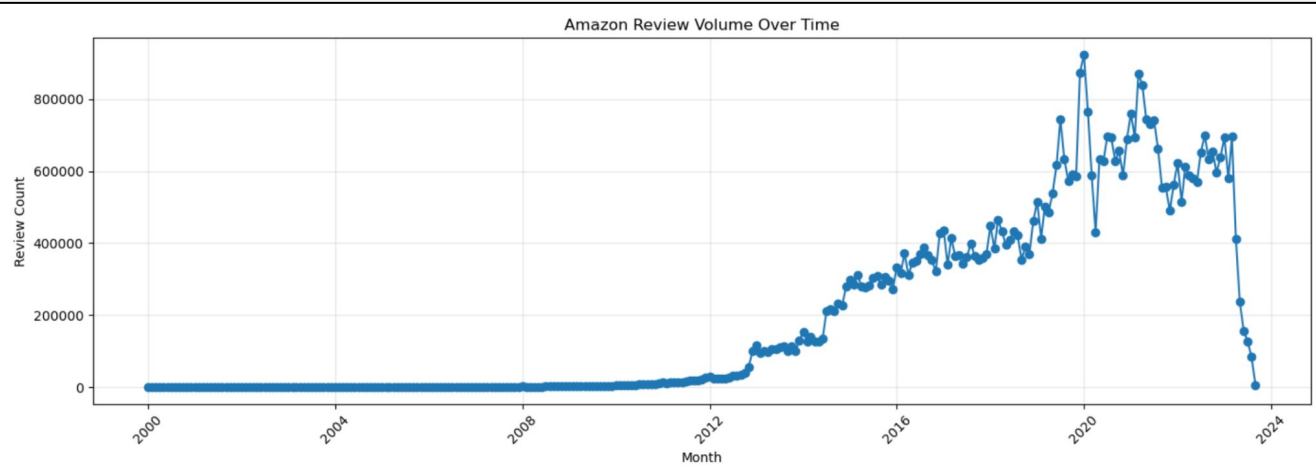
Main Category	Count
All Beauty	16494786
Cell Phones/accessories	14925782
Automotive	13590960
Personal Care	1881180
Electronics	1592778

The dataset spans over 20 product categories, with strong concentration in a few major groups. All Beauty and Cell Phones & Accessories dominate the dataset, each containing tens of millions of reviews.



The cleaned dataset shows a right-skewed rating pattern, with 5-star reviews making up the majority of all ratings.

Amazon Review volume Over Time (data timeline)



Spikes caused by real activities/events

- There are several large spikes align with real Amazon retail events:
 - Prime Day (July)
 - Black Friday / Cyber Monday (November)
 - Holiday shopping periods
 - COVID-era online shopping surge (2020–2021)
- These confirm that the spikes represent real consumer behavior, not noise or data errors.

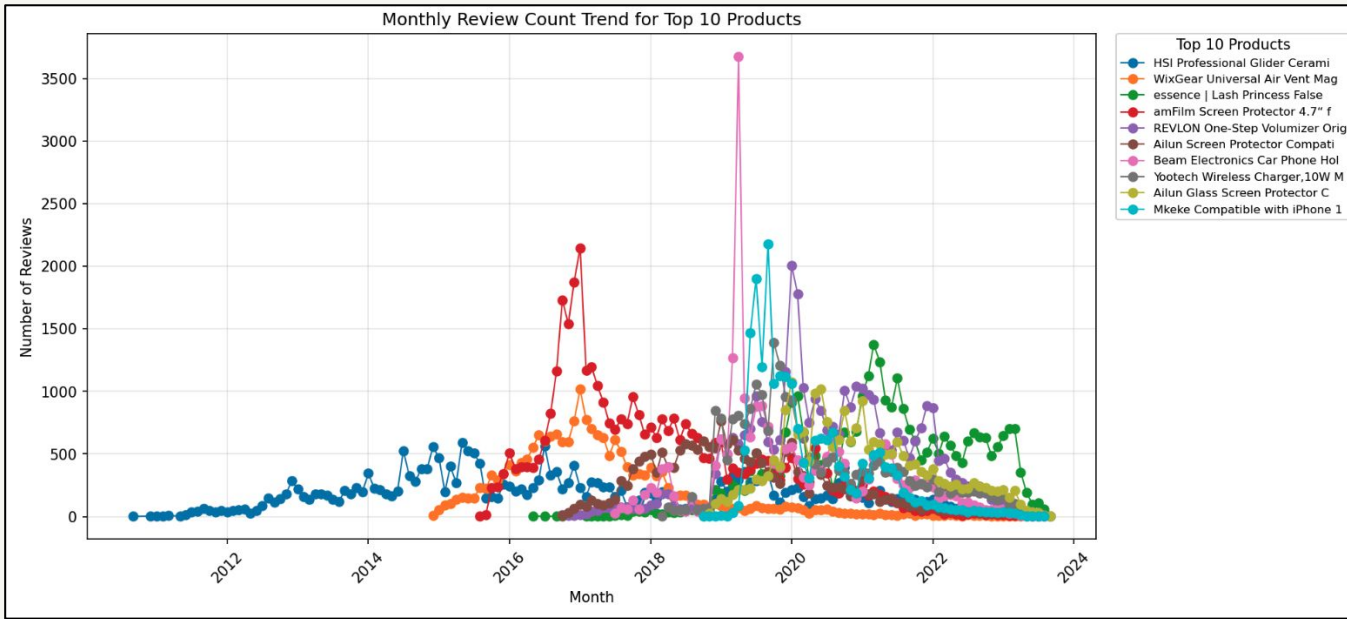
Outliers

Data overall timeline, significant peaks and valleys and data collection gaps

- The dataset spans 2000–2023, but meaningful review activity begins around 2009.
- From 2011 through 2021, review volume steadily increases with noticeable seasonal rises and dips.
- Major peaks appear between 2018–2021, reflecting very high monthly review activity.
- 2000–2008 shows extremely low volume, demonstrating that 2000–2008 is an incomplete historical archive
- Late 2023 shows an abrupt drop because the dataset ends mid-year, not because reviews actually decreased.
- No internal gaps appear during the main activity window (2012–2022).

- Removed structural outliers such as invalid timestamps, extremely short/empty reviews, duplicate entries, unrealistic product prices, and months with near-zero review counts (early sparse years and incomplete 2023 months).
- High peaks in 2019–2021 are not outliers, they are event-driven increases.

Top 10 Most-Reviewed Products & Their Review Trends



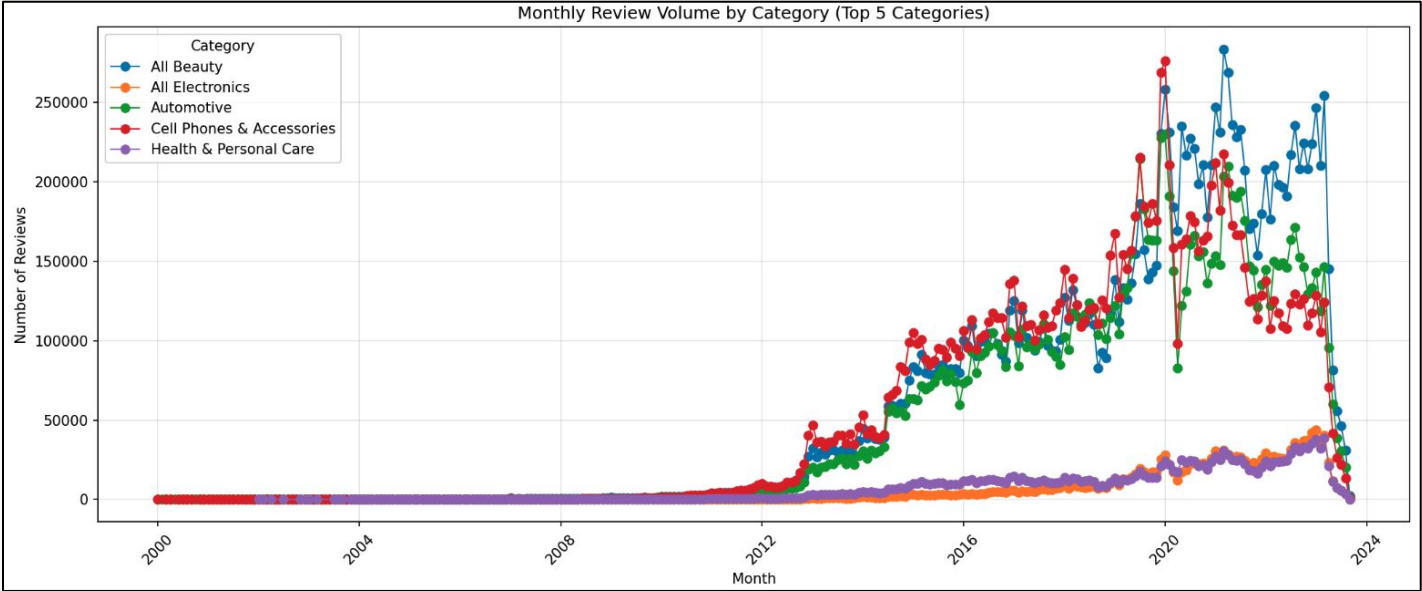
The dataset's most-reviewed products are dominated by phone accessories, beauty items, and car mounts, including:

1. HSI Professional Glider Ceramic Flat Iron
2. WixGear Universal Air Vent Magnetic Car Mount
3. essence | Lash Princess False Lash Effect Mascara
4. amFilm Screen Protector (4.7")
5. REVLON One-Step Volumizer Original
6. Ailun Screen Protector Compatible with iPhone 11 / XR
7. Beam Electronics Car Phone Holder Mount
8. Yootech Wireless Charger, 10W Max
9. Ailun Glass Screen Protector Compatible with iPhone 12 / 12 Pro
10. Mkeke Compatible with iPhone 11 Case

Top product review trends over time

- Each top product shows a "life cycle": reviews ramp up quickly once the item gains traction, then slowly decline as the product becomes older or replaced by newer versions.
- Screen protectors, chargers, and other fast-moving accessories show the largest and sharpest review spikes, because their demand jumps immediately after new phone model launches.
- Many spikes line up with major retail moments such as new iPhone/Samsung releases, the November–December holiday season, and Amazon Prime Day events.
- Some products experience brief viral popularity, leading to short-lived surges in reviews before stabilizing again.
- Starting around 2021, nearly all top products show a notable drop in review activity, consistent with the overall dataset trend.

Review Volume Vary Across Product Categories



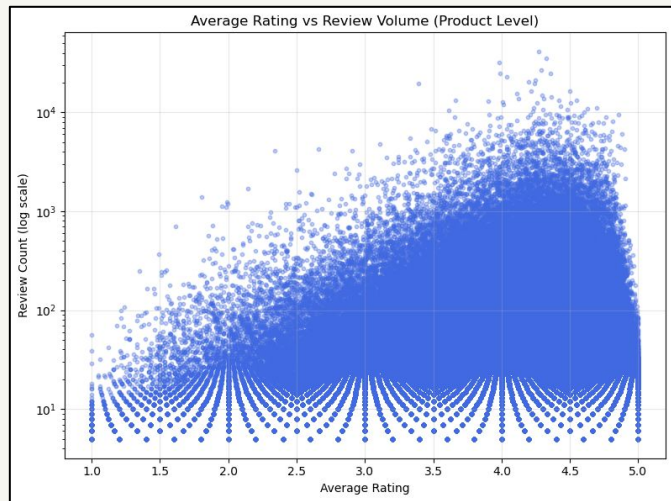
Overall Category Patterns

- Beauty, Cell Phones & Accessories, and Automotive dominate review volume, each reaching 100k–250k+ monthly reviews at peak.
- Electronics and Health & Personal Care grow more slowly but show steady, consistent activity over time.

Summary: Category Review Patterns & Seasonal Spikes

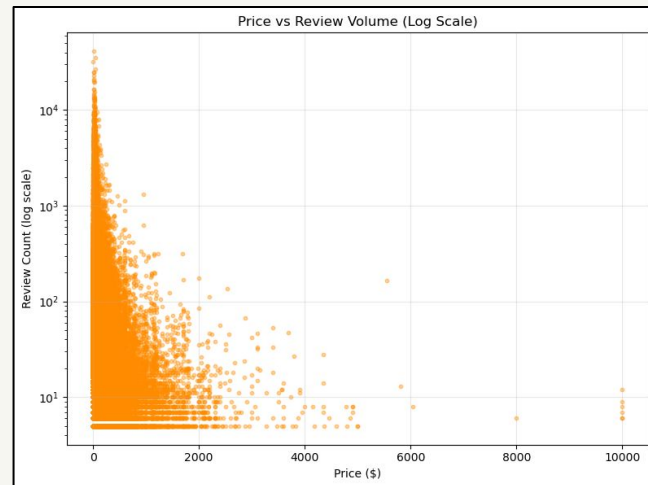
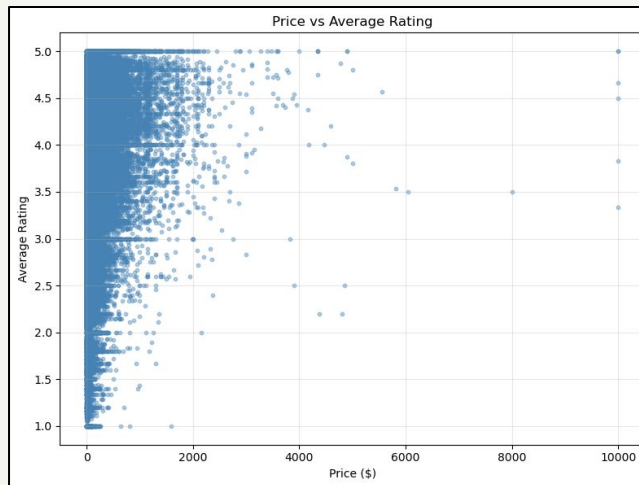
- Cell Phones & Accessories shows the biggest spikes, usually around new iPhone/Samsung launches (Sept–Oct), holiday shopping (Nov–Dec), and Prime Day (July).
- Beauty also jumps during holiday gifting seasons and large promotional events.
- Automotive and Electronics show smaller, steady seasonal lifts, especially toward year-end.
- Overall, review activity is highly event-driven, with Beauty and tech accessories showing the most volatility, while Health & Personal Care and Electronics remain more stable.
- All categories experience a decline after 2021, likely tied to Amazon review policy changes and general market saturation.

What Drives Review Volume Across Products?

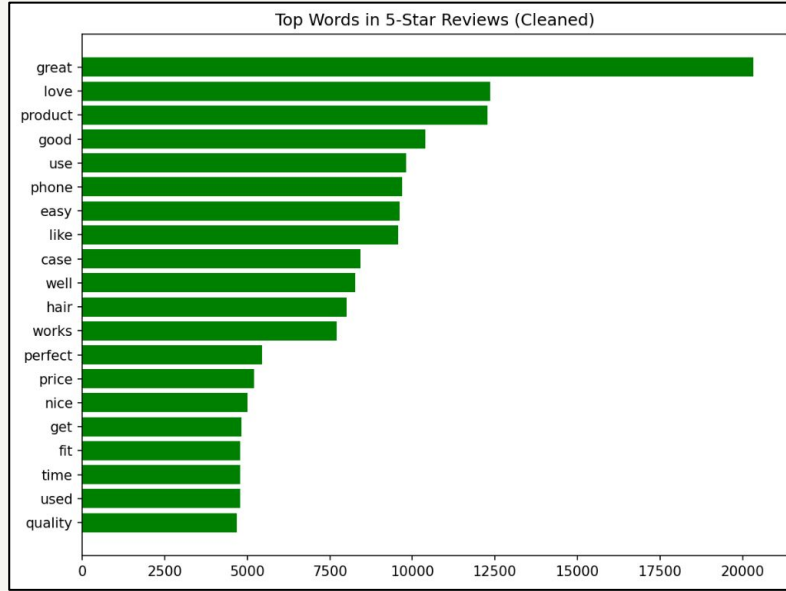


- High ratings does not mean high review volume
- Products with 4.9–5.0 ratings does not have the most reviews. The largest review volumes cluster around 4.3–4.6
- This pattern suggests that extremely high ratings may reflect small, loyal reviewer bases, while large-volume products naturally accumulate more mixed reviews.
- Very high-rated products (4.8–5.0) often receive fewer reviews than those around 4.4–4.6, suggesting that the “perfect rating” items are often niche or newer, rather than mass-market hits.
- The conclusion is products with good but not perfect ratings attract the largest audiences.

- There is no strong correlation between price and rating.
- Both low-cost and high-end products spread across the entire rating spectrum.
- Expensive products tend to have fewer total reviews, likely because fewer customers purchase them
- this shows that consumer satisfaction is not tied to price

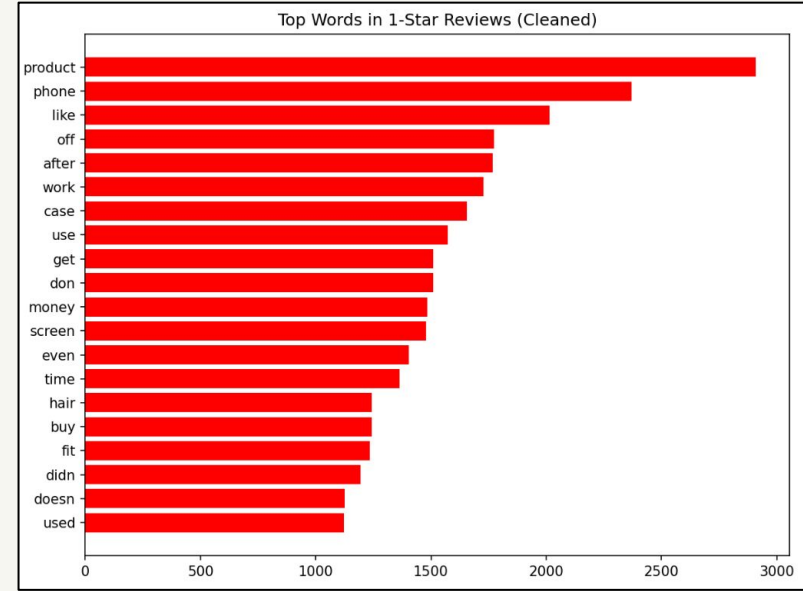


The most common words or phrases in 5-star vs. 1-star reviews



5-Star Reviews — What Positive Customers Emphasize

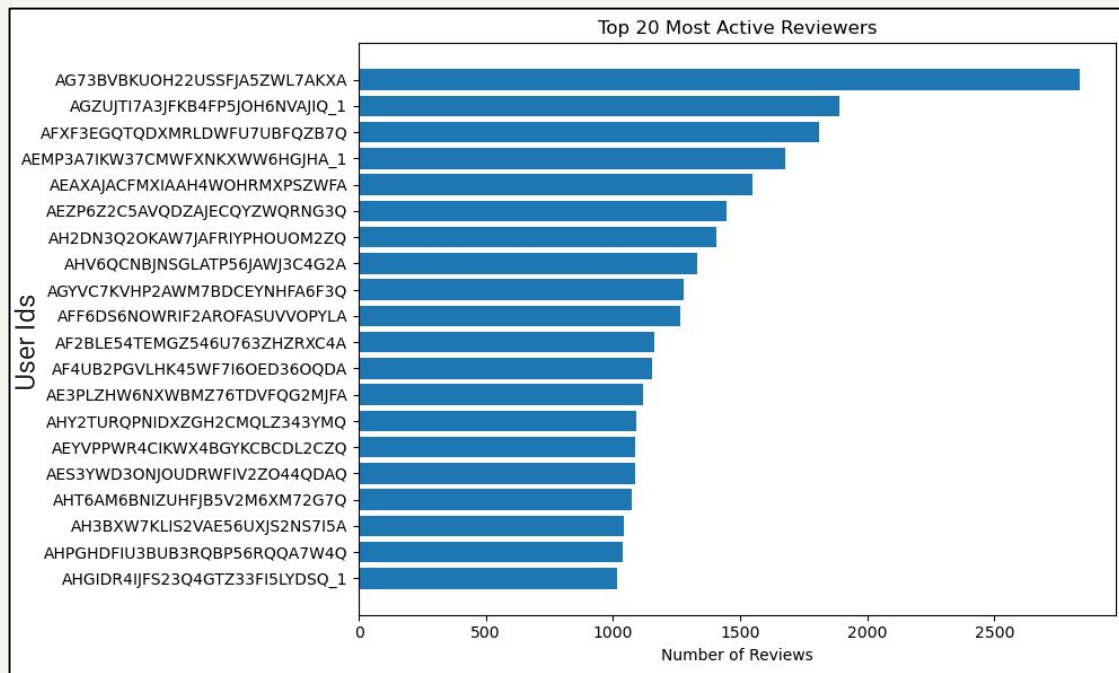
- Strong positive sentiment dominates (“great,” “love,” “good,” “easy”), indicating satisfaction with quality and experience.
- Words like **“product,” “use,” “phone,” “case”** suggest that positive reviews frequently describe practicality and everyday usefulness.
- Frequent mentions of **“perfect,” “nice,” “fit,” “quality”** reflect that customers highlight good value, strong performance, and good fit/function.



1-Star Reviews — What Drives Negative Feedback

- Words like **“product,” “phone,” “like,” “off,” “after,” “work”** indicate that most complaints are functional or performance-related (items not working as expected).
- Terms such as **“get,” “don’t,” “didn’t,” “doesn’t”** reflect dissatisfaction, failures, or unmet expectations.
- Mentions of **“money,” “buy,” “screen,” “fit,” “even”** suggest concerns about value, durability, and incorrect sizing or compatibility.
- Overall, negative reviews contain **more action words** tied to failure modes (“didn’t work,” “doesn’t fit”), showing frustration and problem descriptions.

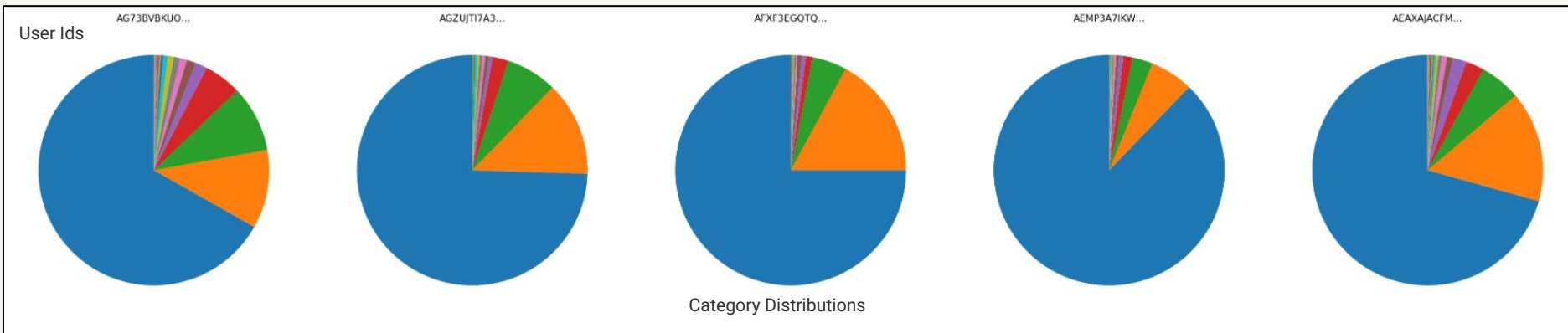
Most active reviewers



- Review activity is highly skewed, with a small cluster of “super-reviewers” dominating total contributions.
- The #1 reviewer posted 2,836 reviews, nearly 40% more than the second-most active reviewer.
- The top 5 reviewers each write 1,800+ reviews, far above the typical user.
- The top 20 reviewers all exceed 1,200 reviews, reinforcing a strong long-tail distribution.
- This concentrated behavior suggests a small user group heavily influences product ratings, sentiment profiles, and category visibility.

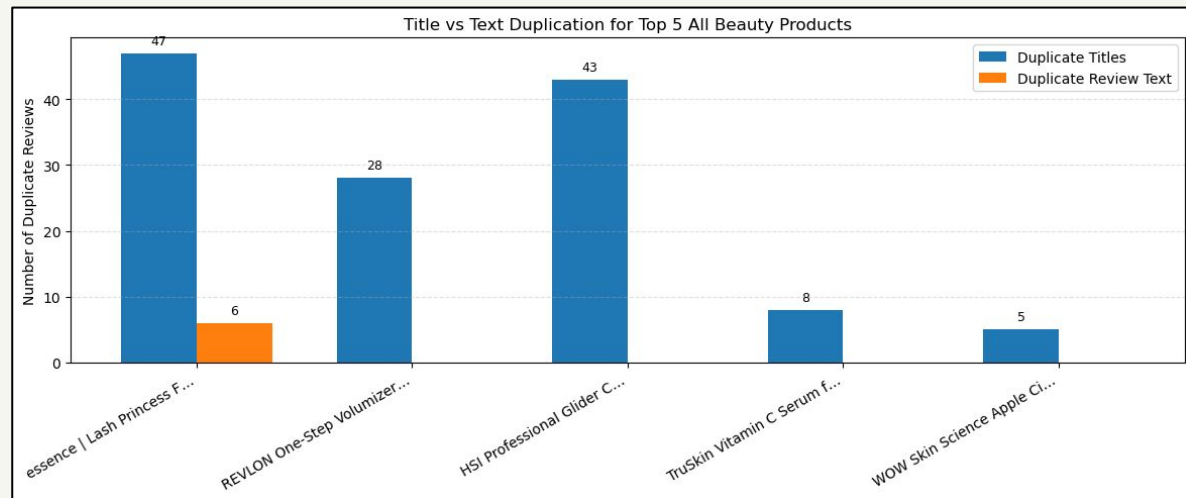
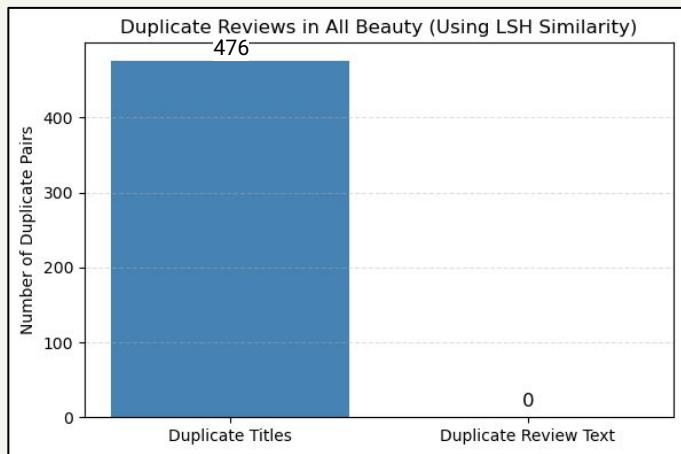
Figure: Top 20 reviewers by total review volume, highlighting the dominance of a small group of high-frequency contributors (“super-reviewers”).

Top Reviewers Category Diversity



- Across the top 5 most active reviewers, posting behavior is highly concentrated in a single category
- Each reviewer writes 60–85% of their reviews in their dominant category (e.g., *All Beauty*, *Premium Beauty*, or *Cell Phones & Accessories*).
- Secondary categories represent smaller slices (5–15%), and the remaining categories make up only marginal contributions.
- This indicates that heavy reviewers are specialists, not generalists—they tend to repeatedly review products in the same category rather than exploring across the catalog.
- As a result, their review patterns likely reflect deep familiarity with a specific product domain rather than broad marketplace usage.

“Review Titles” and “Review Texts” Duplication

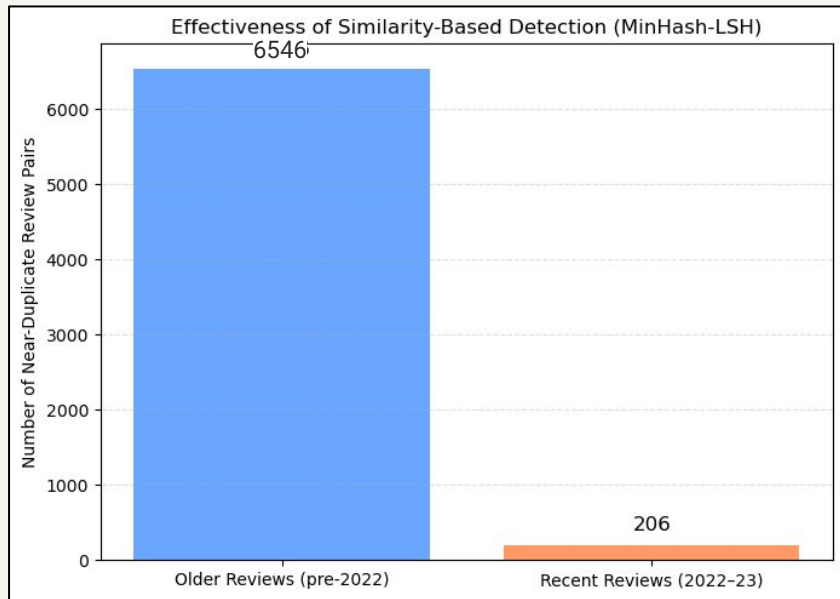


- Using a 1% sample and a 0.2 threshold, duplicate review titles are present, but duplicate review texts are nearly nonexistent in the All Beauty category.
- Using MinHash-LSH similarity, we identified **476 pairs of duplicated or near-identical titles**, indicating many customers tend to reuse similar short phrases like “love this,” “great product,” or format-based titles.

Across the five most-reviewed products, **title duplication remains common**, while **text duplication stays extremely low**:

- **essence | Lash Princess Mascara**: 47 duplicated titles, but only 6 duplicate texts.
- **REVLON One-Step Volumizer Original**: 28 duplicated titles, 0 duplicate texts.
- **HSI Professional Glider Ceramic Flat Iron**: 43 duplicated titles, 0 duplicate texts.
- **TruSkin Vitamin C Serum**: 8 duplicated titles, 0 duplicate texts.
- **WOW Skin Science Apple Cider Vinegar Shampoo**: 5 duplicated titles, 0 duplicate texts.

The effectiveness of similarity-based detection on recent reviews (e.g., 2022–2023) versus older reviews (pre-2022)



Using a 5% sample, older reviews (pre-2022) contain **6,546** near-duplicates, while recent reviews (2022–23) contain only **206**.

Older reviews show strong patterns of repeated or copy-pasted content, likely due to:

- Early spam or bot-like review behavior
- Sellers reusing templated text across products
- Less strict Amazon moderation in the 2010–2018 era

Recent reviews are significantly more diverse, meaning:

- Amazon's spam detection and anti-manipulation policies have improved.
- Content is generated more individually (or more cleverly if AI-assisted).
- There is less obvious repetition for LSH to detect.

Similarity-based methods (MinHash-LSH) detect far more near-duplicate reviews in older data than in recent data.

Conclusions and actionable recommendations

Conclusions

- Review activity is highly uneven, a small cluster of super-reviewers dominates review volume, which can distort product sentiment and visibility.
- High review volume does not equal high satisfaction; the most-reviewed products generally sit in the 4.3–4.6 range, not the extremes.
- Categories like Beauty, Cell Phones, Accessories consistently drive the largest share of reviews.
- Text patterns show clear contrasts: 5-star reviews highlight simplicity and quality, while 1-star reviews focus on specific problems.
- Similarity patterns suggest older reviews contain more templated/spam-like behavior, while newer reviews show more diverse structures likely **shaped by modern moderation and AI-era content.**

Actionable Recommendations

- **Prioritize monitoring high-volume products and users** to detect rating shifts early, since these items influence marketplace perception the most.
- **Apply category-specific review screening**, especially in Beauty and Cell Phones where spikes frequently occur and review manipulation may be more impactful.
- **Use similarity-based tools (MinHash/LSH/SimHash) primarily on older review periods**, where template-style duplication is still detectable; adjust thresholds for recent reviews due to more diverse AI-generated text.
- **Prioritize quality improvements on recurring 1-star keywords** (e.g., “didn’t work,” “didn’t fit,” “broken”), which consistently drive negative sentiment across top products.
- **Boost visibility for strong-rating but low-volume products**, which may perform well but remain overshadowed by mass-reviewed items.
- Adjust similarity thresholds for modern **AI-generated reviews**, which tend to be more syntactically diverse but semantically repetitive.
- **Flag repetitive title structures** as potential indicators of automated or coordinated review activity.